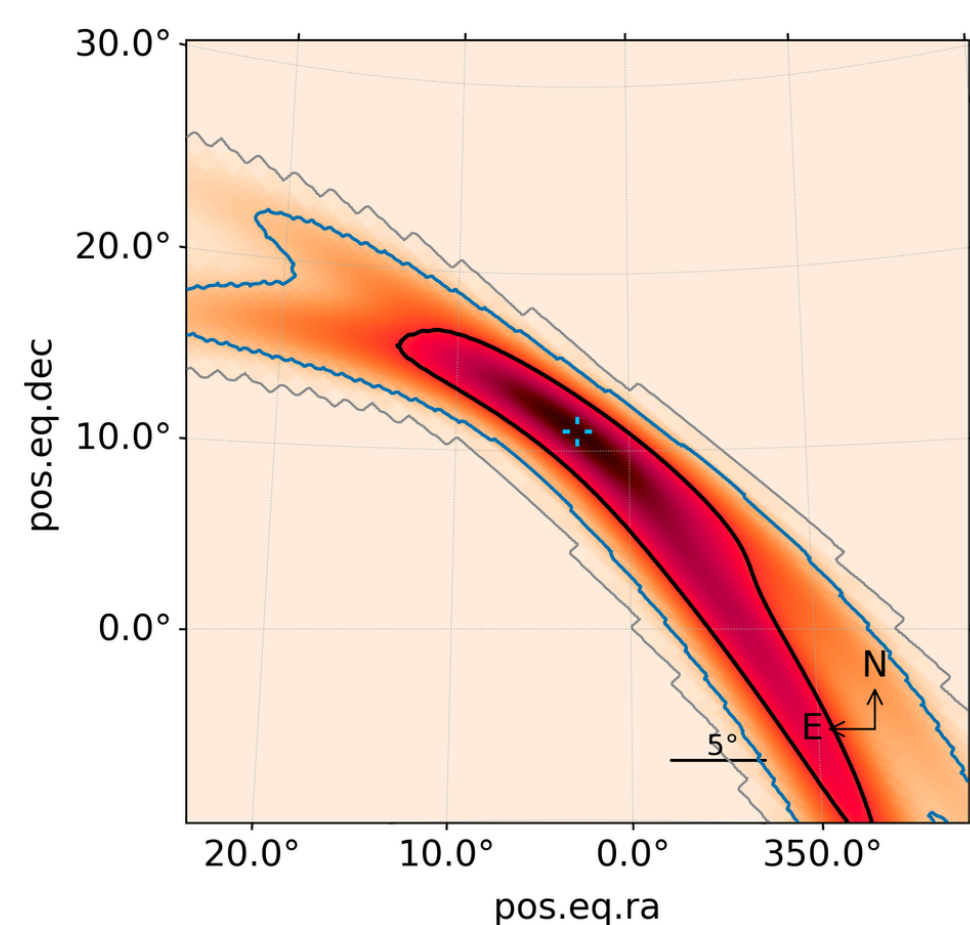
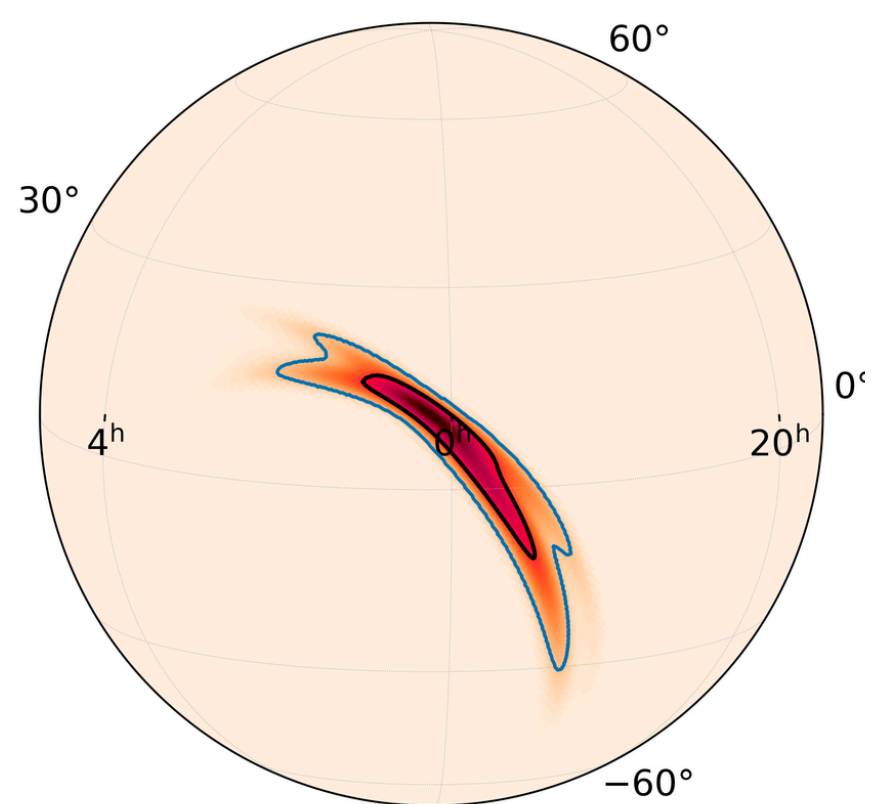


HEALPix Skymap + Distance (GraceDB)



a) TEGLON database

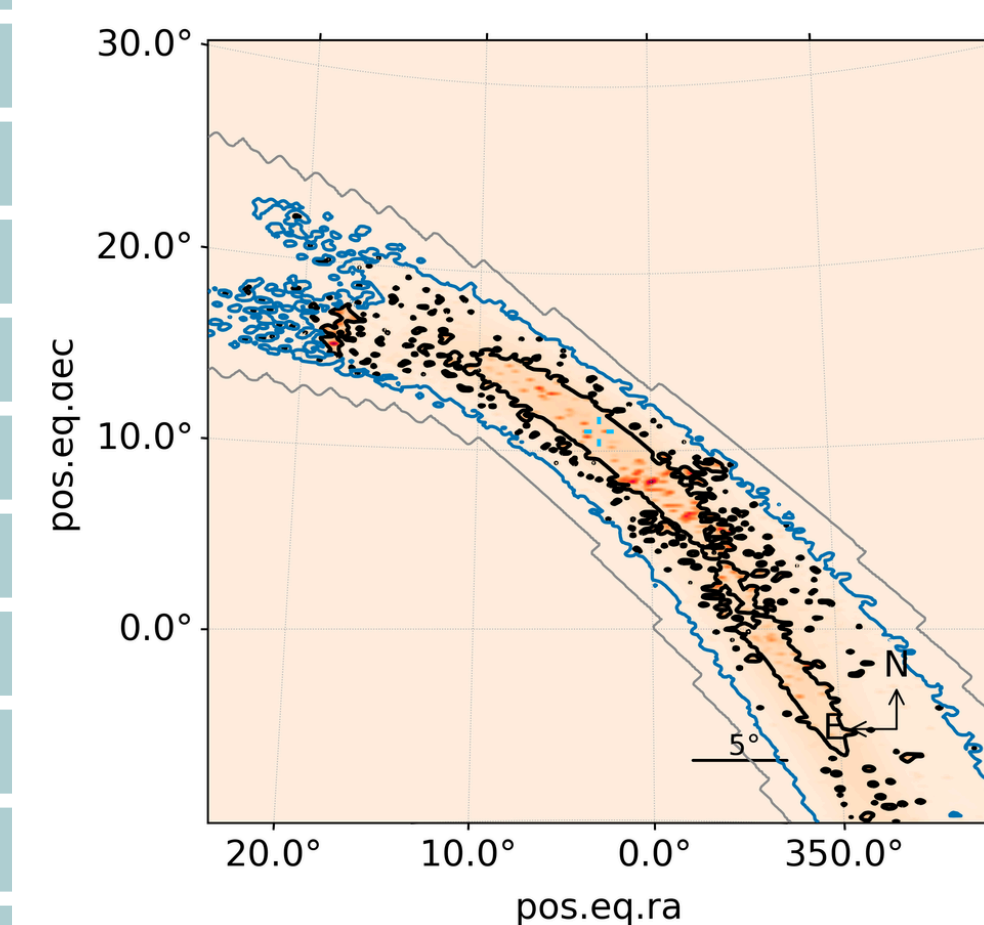
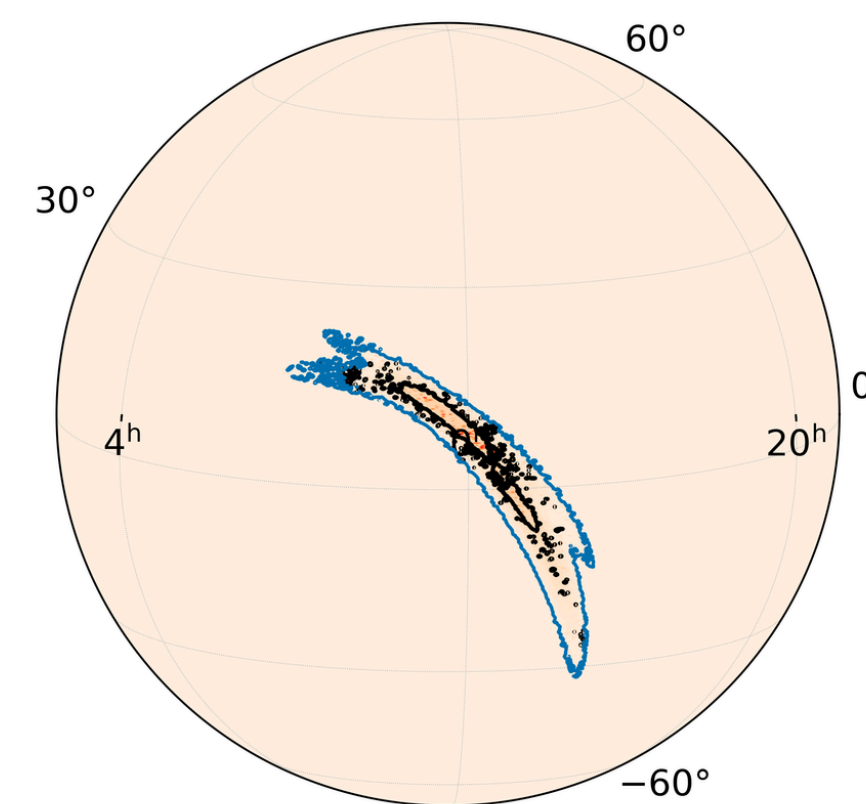
HEALPix grid

Galaxy Table
(RA, DEC, distance)

Detector + Bands
(FOV grids, Filters)



Galaxy-reweighted 4D probability map



b.1) Observation Planning Mode

Telescope
/Detectors

FOV tiling &
probability-per-
area ranking

Galaxy-
reweighted 4D
probability Map

b.2) Analysis Mode

Transient Model
(simulations)

Observed Tiles
(RA, DEC, limiting
magnitude, filter,
time)

Model Efficiency
Calculation

Per-telescope
ranked tile list

Detection
efficiency
(Kilonova Model)

Science output

Galaxy Catalog

GW Data

Scripts

Observed &
simulated data